

DECREASE AND CONQUER



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MOTIVATION

- Why study algorithm design techniques?
- From brute force to smarter strategies



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WHAT IS DECREASE-AND-CONQUER?

- **Solves a problem by:**
 - Reducing problem size
 - Solving a smaller version
 - Extending the solution
- **Only one subproblem is created**





CORE IDEA

- Problem of size n
- Reduce to size:
 - $n - 1$, $n - k$, or n / k
- Build solution from smaller input





GENERAL ALGORITHM STRUCTURE

- If problem is small:
 - solve directly
- Else:
 - reduce problem size
 - solve smaller problem
 - extend solution





TYPES OF DECREASE-AND-CONQUER

- Decrease by a constant
- Decrease by a constant factor
- Variable-size decrease





DECREASE BY A CONSTANT

- Problem size reduced by 1 or k
- Examples:
 - Linear Search
 - Insertion Sort





EXAMPLE: INSERTION SORT

- Sort first $n - 1$ elements
- Insert the n th element correctly





INSERTION SORT COMPLEXITY

- Best case: $O(n)$
- Worst case: $O(n^2)$





DECREASE BY A CONSTANT FACTOR

- Size reduced by a fraction (e.g., $n/2$)
- Much more efficient
- **Example:**
 - Binary Search





EXAMPLE: BINARY SEARCH

- Works on sorted data
- Compare middle element
- Eliminate half of the search space





BINARY SEARCH COMPLEXITY

- Time Complexity: $O(\log n)$
- Space Complexity: $O(1)$ (iterative)



VARIABLE-SIZE DECREASE

- Reduction size depends on input
- Examples:
 - Euclid's Algorithm (GCD)



EXAMPLE: EUCLID'S ALGORITHM

- $\text{GCD}(a, b)$:
- if $b == 0$ return a
- else return $\text{GCD}(b, a \bmod b)$





COMPARISON WITH DIVIDE-AND-CONQUER

- **Decrease-and-Conquer**

- One subproblem
- Simple combine step
- Often linear recursion

- **Divide-and-Conquer**

- Multiple subproblems
- Complex combining
- Branching recursion



ADVANTAGES

- Can be slower than divide-and-conquer
- Not always optimal for large datasets





WHEN TO USE DECREASE-AND-CONQUER

- Problem depends on smaller instances
- Simple extension step
- Ordered or incremental data





SUMMARY

- Reduces problem to one smaller instance
- Three main types
- Widely used in searching, sorting, and math algorithms



END



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